

## FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10

*Student Assessment Document*

### FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10: STATISTICS 1 (3.1)

#### Student Assessment Document

|              |       |
|--------------|-------|
| Student Name | _____ |
| Class        | _____ |
| Date         | _____ |

#### INSTRUCTIONS FOR STUDENTS

Mr Otieno, the principal of Lakeview Senior School in Kisumu, has money for only one option: a school bus route, a bicycle shed or a second lunch serving. In this Final Explanation you will use everything you learned about Grade 10 Green's travel data to answer the driving question: How can we turn a messy list of data from our school into clear evidence that helps our principal make a fair decision? Write your answers in the space under each prompt. Use numbers from the data to back up every claim, give units where needed (learners, seconds), and write the mean as 'mean'. You may use a calculator and a protractor if you need them. Remember that all the data come from one class only.

### Part 1: From a messy list to a frequency table

In Lesson 1, three learners disagreed. Atieno said 'Almost everyone walks', Brian said 'Most of us come by matatu', and Zawadi said 'The bus is always full'. Grade 10 Green collected 40 responses in learner order, but the raw list did not settle the argument.

Here is the frequency table of the same 40 responses:

| Travel mode | Frequency |
|-------------|-----------|
| Walk        | 14        |
| Matatu      | 12        |
| Boda boda   | 4         |
| Bicycle     | 3         |
| Private car | 3         |
| School bus  | 4         |

(a) Explain why the raw list of 40 responses could not settle the claims, but the frequency table can. (b) Work out the percentage for Walk and for Matatu, and the percentage for the two groups together. (c) Use your numbers to decide whether Atieno and Brian are right, and explain what the class data can and cannot tell us about Zawadi's claim.

*Write your answer here:*

## Part 2: Choosing a typical value

Grade 10 Green also counted the number of siblings each learner has:

| Siblings | 0 | 1 | 2  | 3 | 4 | 5 |
|----------|---|---|----|---|---|---|
| Learners | 5 | 9 | 12 | 8 | 4 | 2 |

An Eldoret club group of 7 runners had these 800 m training times in seconds: 104, 105, 105, 106, 107, 108, 130 (the runner with 130 s had a cold).

(a) Find the mean, median and mode of the siblings data. (b) Find the mean, median and mode of the runner times, giving the mean to 1 decimal place. (c) Why do the three measures agree for siblings but differ for the runner times? Which measure would you use for the runner times, and why? (d) Which measure can you use for the travel mode data, and what is its value?

*Write your answer here:*

### Part 3: Choosing and checking graphs

Mr Kamau showed two bar graphs of the same travel data. On one, the frequency axis started at 0. On the other, it started at 2.

(a) Which type of graph is best for the travel data, and which is best for Mama Achieng's mandazi sales at Kibuye market (Mon 40, Tue 45, Wed 42, Thu 50, Fri 65, Sat 30, Sun 20)? Give a reason for each. (b) On the graph where the axis starts at 2, the Walk bar is 12 units tall and the Bicycle bar is 1 unit tall. What does it make walking look like compared with cycling? What is the true comparison from the data? (c) List four things you would check before trusting a graph.

*Write your answer here:*

#### Part 4: Pie chart angles

You drew a pie chart of the 40 travel responses.

**(a) How many degrees does each learner represent? Show your working. (b) Calculate the angle for Walk and for Matatu. (c) Give one advantage of the pie chart and one advantage of the bar graph for the travel data. (d) What does the pie chart show about the bicycle group, and what does this mean for the bicycle shed?**

*Write your answer here:*

### Part 5: Your Final Explanation for Mr Otieno

Write a short report (about 8 to 10 sentences) for Mr Otieno that answers the driving question: How can we turn a messy list of data from our school into clear evidence that helps our principal make a fair decision?

In your report: (1) describe the steps that turn the raw list into evidence; (2) say what the evidence shows about the three claims; (3) give your advice about the bicycle shed, the school bus route and the second lunch serving; (4) explain one limit of the data and how Mr Otieno should deal with it.

*Write your answer here:*

### RUBRIC

| Criterion                                   | Excellent (4)                                                                                                                     | Proficient (3)                                                                                | Developing (1–2)                                                               |
|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| Organising data and calculating from tables | All percentages, angles, means and medians are correct, with clear working and the frequencies checked against the total.         | Most calculations are correct with some working shown; one small slip in rounding or working. | Several calculations are incorrect or have no working; totals are not checked. |
| Choosing measures and graphs                | Chooses mean, median, mode and graph types correctly for each data type, and explains using the outlier and the categorical data. | Chooses most measures and graphs correctly with partial reasons.                              | Choices are unclear or not linked to the type of data.                         |

|                                           |                                                                                                                                                        |                                                                                          |                                                                                           |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Evidence-based reasoning about the claims | Uses exact numbers (14 of 40, 35%; 12 of 40, 30%; 65%) to judge each claim, and correctly says the class count cannot test Zawadi's bus claim.         | Judges the claims with some numbers; the bus claim is mentioned but not fully explained. | Gives opinions without numbers, or states the claims are right or wrong without evidence. |
| Critical reading of graphs                | Explains the truncated axis effect (12 times against a true ratio of about 4.7) and lists several correct checks for an honest graph.                  | Explains that the axis can mislead and lists some checks.                                | Does not recognise how a graph can mislead.                                               |
| Final recommendation and communication    | Clear, honest report that links the steps from raw list to evidence, advises no shed, recommends whole-school data, and states the limit of one class. | Gives a clear recommendation but misses one step or the limit of the sample.             | Recommendation is unclear or is not supported by the data.                                |

## ATTRIBUTION AND LICENSING

**Curriculum Alignment:** Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

**Content & Pedagogy:** Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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